EduVision Analytics System

Executive Architecture & Dashboard Story Document

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1. Executive Summary & Project Purpose

Strategic Mission

The **EduVision Executive Higher Education Analytics Platform** is designed to bridge raw global university performance data with actionable executive insights for academic leaders, policy makers, and prospective students. By synthesizing multifaceted global rankings into intuitive interactive views, EduVision empowers stakeholders to benchmark institutional performance, assess research impact, evaluate student demographics, and formulate data-driven national strategies.

Data Pipeline Architecture

The system utilizes a structured end-to-end data processing workflow designed to maintain high analytical integrity and fast visual render times:

1. **Ingestion:** Raw Kaggle global higher education datasets covering university performance, student body stats, and research metrics.
2. **Transformation:** Data cleaning, normalization, missing value imputation, and feature engineering executed via Python in Jupyter Notebooks.
3. **Data Modeling:** Modeled into an optimized **Star Schema** consisting of 2 Conformed Dimension Tables and 4 Specialized Fact Tables.
4. **Visual Analytics:** Interactive executive dashboards constructed and deployed in Tableau Desktop and Tableau Public.

[Raw Kaggle Data] ──> [Python / Jupyter Processing] ──> [Star Schema Model] ──> [Tableau Analytics Platform]

Star Schema Data Architecture

Table Name	Entity Type	Primary Key / Foreign Keys	Description
`dim_country`	Conformed Dimension	`country_id`	Geographic attributes, region classifications, and regional GDP metrics.
`dim_university`	Conformed Dimension	`university_id`, `country_id`	Core metadata for academic institutions, foundation dates, and governance models.

Table Name	Entity Type	Primary Key / Foreign Keys	Description
`fact_university_overview`	Fact Table	`overview_id`, `university_id`, `year`	High-level performance indicators, overall rank, and composite scoring metrics.
`fact_research_analytics`	Fact Table	`research_id`, `university_id`, `year`	Scholarly output details, citation counts, research income, and impact ratios.
`fact_student_analytics`	Fact Table	`student_id`, `university_id`, `year`	Enrolment metrics, international student proportions, and faculty-to-student ratios.
`fact_country_comparison`	Fact Table	`comparison_id`, `country_id`, `year`	Aggregated national-level higher education benchmarks and performance averages.

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2. User Personas & Strategic Journeys

Persona	Role & Focus	Strategic Need	Core Journey
Persona 1	**University Chancellor / Executive Leadership**	Institutional standing, national rank, and overall institutional prestige.	Tracks overarching annual progress, identifies national rank gaps, and communicates prestige metrics to governing boards.
Persona 2	**Vice-Chancellor of Research & Innovation**	Scholarly output, citation density, and academic productivity vs. overall rank.	Analyzes output efficiency, identifies underperforming research sectors, and allocates grant funding.
Persona 3	**Admissions & Student Affairs Director**	International mobility, student body diversity, and teaching environment ratios.	Evaluates global talent recruitment drives and monitors staff workload ratios to preserve educational quality.
Persona 4	**Higher Education Policy Analyst**	Systemic country performance, regional strength, and cross-border benchmarking.	Compares national performance against competing higher education systems to shape funding policy.

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3. Dashboard-by-Dashboard Story Narratives

Dashboard 1: University Overview (Global & National Standing)

- **Executive Narrative:** *"Where do we stand globally and nationally?"*
- **Core Metrics & KPIs:** Total Monitored Institutions, Top-Ranked University Count, Average Overall Score.
- **Visual Journey & Interactions:**
    - **Global Distribution Map:** Interactive map enabling top-level geographical filtering across all dashboard charts upon country selection.
    - **Top University Leaderboard:** Dynamic listing displaying the Top 10 institutions based on selected filters, adjusting seamlessly between global and country contexts.
    - **Score Distribution Charts:** Visualizes the concentration of overall scores across tier bands to evaluate competitive density.
- **Key Insights:** Top-tier universities demonstrate high performance consistency across academic reputation and composite scoring, whereas mid-tier institutions exhibit higher score volatility dependent on national funding models.

Dashboard 2: Research Analytics (Citations & Academic Output)

- **Executive Narrative:** *"How effectively is our institution driving impactful research?"*
- **Core Metrics & KPIs:** Average Research Score, Average Citation Rating, High-Impact Output Ratios.
- **Visual Journey & Interactions:**
    - **Research Score vs. Academic Reputation Scatter Plot:** Evaluates the direct correlation between hard research output and global peer perception.
    - **Citation Leaderboard:** Highlights top-performing research institutions globally and regionally to identify scholarly benchmarks.
- **Key Insights:** High research citation ratings directly drive long-term global brand equity and institutional prestige; however, a lag of 2–3 years is typical between publication spikes and reputation score increases.

Dashboard 3: Student Analytics (Diversity & Learning Environment)

- **Executive Narrative:** *"What does our student body look like, and are faculty resources sufficient?"*
- **Core Metrics & KPIs:** Total Enroled Students, International Student Count, International Student Percentage, Faculty-to-Student Ratio.
- **Visual Journey & Interactions:**
    - **Student Diversity Bar Chart:** Ranks institutions by international student concentration to measure global attractiveness.
    - **Students Per Staff vs. Overall Score:** Evaluates classroom density against academic performance to identify optimal staffing ratios.
    - **Global Student Distribution Map:** Visualizes international student mobility and recruitment corridors.
- **Key Insights:** Institutions that maintain a student-to-staff ratio below 12:1 exhibit significantly higher student satisfaction and academic outcome scores, particularly when paired with an international student body exceeding 20%.

Dashboard 4: Country Comparison (National Benchmarking)

- **Executive Narrative:** *"How does our country's higher education system compare to global peers?"*
- **Core Metrics & KPIs:** Country Institutions Count, National Average Score, Average Research Rating by Country.
- **Visual Journey & Interactions:**
    - **Multi-Country Benchmarking Chart:** Direct side-by-side metric comparison across selected benchmark nations.
    - **Research Impact vs. Overall Score Trend:** Evaluates how national higher education spending correlates with global competitiveness over time.
- **Key Insights:** Pinpoints national systemic strengths and highlights specific areas requiring targeted public capital allocation to build global research competitiveness.

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4. Technical Interactivity & Navigation Rules

To guarantee smooth user flows and prevent visual clutter, the EduVision analytics platform enforces a strict set of interactivity rules across all dashboard views:

- **Global Filters:** Every dashboard features an explicit **Year Selector** and a **Cascading Country Dropdown** configured with *"Only Relevant Values"* to prevent invalid dimension queries.
- **Protected Leaderboards:** Ranking worksheets are isolated from detail filters that would break rank calculations, ensuring that dynamic Top 10 views display absolute top performers rather than top items within an overly restricted slice.
- **Context Filter Logic:** Critical geographic and category filters are added to Context (represented as grey pills in Tableau), ensuring that Top N filtering rules are calculated *after* categorical scope is applied.
- **Cross-Dashboard Action Rules (`Ctrl` + `Shift` + `D`):**
    - **Filter by University ID:** Selecting a specific institution on Dashboard 1 cross-filters performance data across Dashboard 2 (Research) and Dashboard 3 (Student Analytics).
    - **Filter by Country ID:** Clicking a nation on any geographic map triggers a direct drill-through navigation to Dashboard 4 (Country Comparison), passing the selected country context automatically.

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5. Deployment & Live Link

The production build of the EduVision Analytics Platform is deployed on Tableau Public for stakeholder access and evaluation.

- **Live Interactive Platform:** [EduVision Analytics on Tableau Public]
- **Official Data Attribution:** Source: [public.tableau.com]